

3(a)	$Q = It$	C1
	$= 3.3 \times 10^4 \times 2.6 \times 10^{-5}$	A1
	$= 0.86 \text{ C}$	
3(b)	$P = IV$ or $P = VQ / t$ or $V = IR \text{ and } P = V^2/R \text{ or } P = I^2R$	C1
	$P = 3.3 \times 10^4 \times 3.0 \times 10^7$ or $P = (3.0 \times 10^7 \times 0.86) / (2.6 \times 10^{-5})$ or $P = (3.0 \times 10^7)^2 / 910$ or $P = (3.3 \times 10^4)^2 \times 910$ $P = 9.9 \times 10^{11} \text{ (W)}$ $= 990 \text{ GW}$	A1
3(c)(i)	$R = \rho L / A$	C1
	$9.6 = 1.7 \times 10^{-8} \times 95 / \pi r^2$	C1
	$r = 2.3 \times 10^{-4} \text{ m}$	A1
3(c)(ii)	(resistance) decreases by a factor of four	A1

Question	Answer	Marks
3(d)	$E = \sigma / \varepsilon$	C1
	$x = \sigma L / E$ $= 1.9 \times 10^6 \times 0.12 / (1.3 \times 10^{11})$	C1
	$= 1.8 \times 10^{-6} \text{ m}$	A1